

ANTECEDENTS OF INDIVIDUALS' INTERTEAM COORDINATION: BROAD FUNCTIONAL EXPERIENCES AS A MIXED BLESSING

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This manuscript investigates the role of individual team members' breadth of functional experience for their interteam coordination behavior. Integrating personal construct and social identity theories, we examine interpersonal cognitive complexity as a mediating variable and organizational identification as a moderator. We test our hypotheses across two independent field studies, comprising an international peace support training mission (Study 1) and a municipality administration (Study 2). Corroborating our predictions, interpersonal cognitive complexity appeared as a conditional mediating variable that can translate an individual's breadth of functional experience into interteam coordination. The strength and direction of this indirect relationship, however, depended on the individual's identification with the organization as a whole. Moreover, on the team level of analysis, we found members' overall interteam coordination to positively relate with team performance in Study 2. All in all, this paper advances new knowledge on the antecedents, mechanisms, contingency factors, and team-level consequences of members' boundary spanning.

Organizational teams do not operate in a vacuum. To accomplish complex tasks and effectively respond to environmental contingencies, team members must engage in boundary spanning efforts to organize task-related issues with other teams (i.e., interteam coordination; Marrone, 2010). Indeed, members' interteam coordination has been shown to increase both team and overall organizational performance (e.g., Ancona & Caldwell, 1992; Drach-Zahavy, 2011; Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). At the same time, interteam coordination may pose considerable difficulties and challenges for individual members, such as overcoming language, goal, and thought-world dif-

ferences between teams (Dougherty, 1992). Uncovering factors that promote or prevent interteam coordination has therefore become an important research agenda for management scholars (Joshi, Pandey, & Han, 2009; Marrone, 2010).

One key factor theorists have identified as important for individual members' interteam coordination is the extent to which members have accumulated work experiences across different functional domains relevant for the organization (i.e., breadth of functional experience; Bunderson, 2003). Scholars have theorized that cross-functional exposure can help individuals to make sense of other teams' distinct functional perspectives and the wider organizational context (Bunderson & Sutcliffe, 2002; Burke & Steensma, 1998; Joshi et al., 2009). Such capacity, in turn, is suggested to enable individuals to bridge their differences with members of external teams, thereby increasing between-team collaboration (Griffin & Hauser, 1996; Maltz, 1997; Raskas

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& Brick, 1992). Some scholars have even gone so far as to argue that moving people across functional boundaries is “one of the most powerful tools that an organization can use” to promote interteam coordination and strengthen boundary spanning between organizational groups (Beechler, S ndergaard, Miller, & Bird, 2004: 129).

Despite this confidence vested in the role of broad functional experiences for individual members’ interteam coordination behavior, direct empirical investigations of this relationship are absent (Joshi et al., 2009). Thus, it remains to be seen whether a positive relation between breadth of functional experience and interteam coordination really exists. In fact, some researchers have argued that broad functional experiences will create “jack[s] of all trades but master[s] of none” (Buyl, Boone, Hendriks, & Matthyssens, 2011: 170), instilling a superficial approach toward multiple functional fields that diminishes individuals’ capacity and/or motivation to thoroughly coordinate complex interteam tasks. Accordingly, empirical studies in adjacent research areas have painted an inconsistent picture. On the department level of analysis, Parry and Song (1993) found a higher percentage of members with multifunctional backgrounds to strengthen integration between different organizational units, whereas Leenders and Wierenga (2002) found cross-functional job rotation to be unrelated with such integration (see also Chimhanzi, 2004; Dawes & Massey, 2005; Maltz & Kohli, 2000). Strong claims for the relevance of broad functional experiences notwithstanding, it therefore remains unclear whether increasing individuals’ respective experiences will, in fact, promote their interteam coordination behavior. To resolve this important problem, we believe it is critical to consider two related issues.

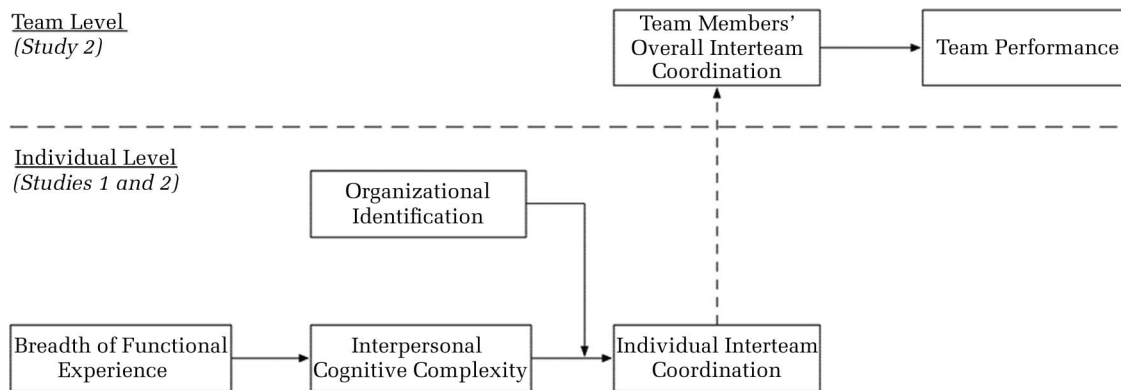
First, it is important to note that broad functional experiences represent demographic background characteristics that are relatively remote from an individual’s day-to-day work behavior (Mark czy, 1997). Hence, it is crucial to examine more proximal, psychological variables that transfer the role of such experiences. As noted before, theorists have emphasized cognitive mechanisms as a key explanation for the role of broad functional experiences (e.g., Griffin & Hauser, 1996; Raskas & Brick, 1992). We therefore draw from personal construct theory as an influential conceptual perspective on the causes and consequences of individuals’ cognition (Kelly, 1955; Walker & Winter, 2007) to understand how broad functional experiences shape an indi-

vidual’s interteam coordination. Specifically, we argue that a broader functional background will increase an individual’s socio-cognitive capacity for interpreting diverse social situations (i.e., interpersonal cognitive complexity; Crockett, 1965). Interpersonal cognitive complexity, in turn, may enable an individual to better deal with between-team situations, and, thus, create the potential for interteam coordination (Hale, 1980; O’Keefe & Sypher, 1981).

Second, to account for inconsistent theoretical predictions and findings on the relationship between breadth of functional experience and interteam coordination, it appears vital to consider moderating factors. Personal construct research suggests that individuals with broad functional experiences may employ the potentials created through interpersonal cognitive complexity for varied ends (Burleson & Caplan, 1998). These potentials may be used, for example, to coordinate interteam tasks, but also to actively avoid the challenges associated with interteam coordination (cf. Bacue & Samter, 2001; Burleson & Denton, 1997). To fully understand the role of broad functional experiences, we therefore believe it is crucial to extend personal construct theory’s purely cognitive scope by incorporating key motivational processes that shape employees’ deployment of their cognitive potential. Social identity theory, a prominent theoretical approach toward intergroup relations, appears particularly useful in this regard, because it highlights motivational factors that specify clear-cut directions for an individual’s social behavior (see Tajfel, 1982; Turner, Hogg, Oakes, Reicher, & Wetherell, 1987). Hence, we integrate personal construct theory with insights from social identity theory, arguing that organizational identification (i.e., a person’s perceived “belongingness” to the overall organization; Mael & Ashforth, 1992) serves as a key contingency factor that influences members’ motivation to utilize their cognitive complexity for or against interteam coordination, and, thus, shapes the indirect linkage between breadth of functional experience and individuals’ respective behavior.

Finally, beyond examining antecedents of individual members’ interteam coordination, we recognize that such behavior can have important implications for a member’s team as a whole (Marrone, Tesluk, & Carson, 2007). Thus, to reiterate the relevance of understanding the origins of interteam coordination, we also investigate the team-level performance consequences that may arise if a work team engages in this type of behavior. The relationships examined in this research are summarized in Figure 1.

FIGURE 1
Conceptual Model



Note: The dashed arrow reflects an aggregation mechanism.

We test this model across two independent studies, comprising (a) an international peace support training mission and (b) a municipality administration. In doing so, our investigation strives to make several contributions. First, it responds to recent calls for more research on the antecedents of interteam coordination, adding to our fundamental understanding of why some individuals span team boundaries to a greater extent than others do (Drach-Zahavy & Somech, 2010; Joshi et al., 2009; Marrone, 2010). In particular, we aim to resolve key ambiguities surrounding the implications of broad functional experiences by demonstrating, in an integrative manner, both how and under what conditions such experiences matter for interteam coordination. Additionally, we underline the importance of interteam coordination by corroborating its team-level performance consequences. And, finally, we strive to realize important theoretical and practical advances. On the one hand, we introduce personal construct theory as a fruitful conceptual framework that prior organizational research has largely neglected, and we illustrate how integrating predictions from personal construct and social identity theory opens up new conceptual perspectives for research on interteam collaboration. On the other hand, our findings may prove valuable for organizations aiming to stimulate interteam coordination, providing new insights on both the selection and management of boundary spanners.

THEORY AND HYPOTHESES DEVELOPMENT

Interteam coordination refers to an individual member's efforts to align his or her team's actions

with other teams in order to manage organization-wide dependencies and workflows (Marks et al., 2005; Mathieu, Marks, & Zaccaro, 2001). Such behavior requires team members to obtain insights into the working of diverse teams, to share information with these teams, and to mutually adjust with external members (DeChurch & Marks, 2006). Scholars have also discussed other behaviors designed to span boundaries between teams, such as searching for information (scouting) and representing one's own team's interests toward management (ambassadorial activities; Marrone, 2010). Importantly, scouting and ambassadorial activities are primarily directed toward promoting one's own team's benefits (e.g., by gaining superior information; Joshi et al., 2009; Marrone, 2010). In fact, such behaviors may even subtract resources from other teams and hamper the overall organization's success (e.g., by transferring assets from other teams toward one's own team; Faraj & Yan, 2009). By contrast, interteam coordination is unique in its focus on the alignment of different teams' task activities to benefit the organization as a whole (DeChurch & Marks, 2006; Smith, Carroll, & Ashford, 1995). Such behavior does not promote one's own team's interests at the expense of other teams, but it explicitly facilitates shared goals (Mathieu et al., 2001). As such, interteam coordination has become a predominant focus of contemporary research on boundary spanning, and scholars have repeatedly demonstrated its distinct relevance for both team and organizational outcomes (Hoegl, Weinkauff, & Gemuenden, 2004; Lanaj, Hollenbeck, Ilgen, Barnes, & Harmon, 2013; Marks et al., 2005).

Breadth of Functional Experience and Interteam Coordination

Personal construct theory as a conceptual foundation. With previous theory emphasizing the role of cognitive mechanisms in the breadth of functional experience–interteam coordination linkage, we draw from the cognition-based perspective provided by personal construct theory (Kelly, 1955) to further explore this notion. Personal construct theory has been widely applied in the fields of communication and psychology (Burleson, 2007; Fransella, 2003), and it is generally viewed as a well-supported approach toward understanding individuals' thinking (Walker & Winter, 2007). The core premise of personal construct theory is that individuals utilize cognitive structures (i.e., personal constructs) to make sense of their environment (Kelly, 1955). Personal constructs consist of abstract characterizations of prior experiences that are stored in individuals' memory and used to interpret subsequent events and derive appropriate reactions (Delia, 1977; Ginsberg, 1990). These personal constructs are malleable; the number of constructs available may expand substantially when an individual confronts novel environments that stimulate the development of new cognitive structures (Buckingham, 1998; Kelly, 1955).

Personal construct theory further suggests that individuals organize their personal constructs into domain-specific subsystems (e.g., referring to furniture, automobiles, business strategies, etc.; Burleson & Caplan, 1998; Ginsberg, 1989). One subsystem that is particularly important for the present purpose relates to the social domain and encompasses the specific set of personal constructs individuals use to "interpret, anticipate, and evaluate the thoughts and behaviors of others" (i.e., interpersonal constructs; Burleson & Samter, 1990: 168). Examples of interpersonal constructs that individuals might use to describe specific social settings or other individuals could be "focused on short-term versus long-term outcomes" and "adjusted versus maladjusted" (Zinkhan & Braunsberger, 2004: 579). Individuals with a highly developed cognitive subsystem in the social domain, comprising a large number of interpersonal constructs, are said to have high *interpersonal cognitive complexity* (Crockett, 1965).

The development of interpersonal constructs is contingent on an individual's experiences within the social domain (Burleson & Waltman, 1988; Kelly, 1955). Studies have, accordingly, illustrated that varied social life experiences (e.g., experiences

in a wide range of business contexts) expand one's interpersonal cognitive complexity (Burleson, 2007; Wicker, 1969). Moreover, scholars have identified such complexity as a fundamental socio-cognitive capacity that critically influences an individual's interpersonal behavior (Burleson & Caplan, 1998; Hale, 1980; O'Keefe & Sypher, 1981). Individuals with high interpersonal cognitive complexity have been shown, for example, to more effectively perform sophisticated social activities (e.g., persuading diverse others; Burleson & Caplan, 1998). As such, personal construct theory—and its predictions regarding interpersonal cognitive complexity in particular—may bear special relevance to explaining the relationship between an individual's breadth of functional experience (as a specific source of diverse social experiences) and interteam coordination (as a specific type of social behavior).¹

Breadth of functional experience and interpersonal cognitive complexity. As noted, personal construct research has shown that interpersonal cognitive complexity can develop substantially with the social experiences individuals encounter throughout their lifetime (Duys & Hedstrom, 2000; Fournier & Payne, 1994; Leitner, 1988). To explicate this linkage, personal construct theory suggests that individuals with direct, in-depth experiences across diverse social settings develop a more elaborate cognitive subsystem to interpret social environments (Leitner, 1988; Wicker, 1969). Such individuals are likely to have repeatedly faced social situations that their existing interpersonal constructs could not accurately represent (Kelly, 1955; Louis, 1980). These out-of-the-ordinary experiences trigger individuals to add novel, differentiated interpersonal constructs to their cognitive subsystem (Fournier & Payne, 1994; Fournier, 1996; Kelly, 1955). Hence, Burleson (2007: 124) argued that interpersonal cognitive complexity is strength-

¹ Scholars have also discussed "integrative" cognitive complexity, referring to an individual's overall tendency to think in advanced ways about a broad range of topics and events (e.g., Suedfeld & Bluck, 1993). Notably, however, research has consistently demonstrated that interpersonal cognitive complexity is more clearly defined and more important for individuals' social behavior, as compared with other types of cognitive complexity (Burleson & Waltman, 1988; O'Keefe & Sypher, 1981). As such, interpersonal cognitive complexity has become a major focus in the personal construct literature (Burleson, 2007). In line with this work, we therefore focus on interpersonal cognitive complexity in the present investigation.

ened to the extent individuals “expose [themselves] to as many different types of people and social experiences as [they] can.”

Transferring these theoretical notions toward an organizational setting, we note that distinct functional work domains typically differ on multiple social dimensions, including norms, language use, competencies, and orientations (Griffin & Hauser, 1996; Homburg & Jensen, 2007; Lawrence & Lorsch, 1967). Members of an engineering department, for example, may primarily focus on technical specifications and production processes, whereas sales personnel may primarily focus on customer relations and turnover targets (Homburg & Jensen, 2007). These functional differences may go along with distinct conceptualizations about how to interact with others (Dougherty, 1992; Prahalad & Bettis, 1986). Over the course of their working life, individuals with experiences across numerous functional domains are therefore directly exposed to multiple social contexts with diverse modes of interaction. As a result, such individuals may develop a greater number of interpersonal constructs, as compared to specialists with work experiences concentrated in few functional areas (Beyer, Chatopadhyay, George, Glick, & Pugliese, 1997; Waller, Huber, & Glick, 1995). Experiences across both engineering and sales departments could, for example, help a person to identify the social dimensions that differentiate these domains, and, accordingly, develop interpersonal constructs that encompass both a production and a customer focus. Consistent with this reasoning, Goodwin and Ziegler (1998) demonstrated, in a study of 220 employees working within various organizations, that individuals with experiences across many different jobs exhibited greater interpersonal cognitive complexity than those with less diverse job experiences.

Hypothesis 1. There is a positive relationship between an individual's breadth of functional experience and his or her interpersonal cognitive complexity.

Interpersonal cognitive complexity and interteam coordination. Personal construct theory further maintains that interpersonal cognitive complexity constitutes an important foundation for individuals' social behavior (Delia, 1977; Kelly, 1955). Specifically, interpersonal constructs are suggested to function as cognitive lenses through which people interpret social situations and make inferences about others (Burlinson & Caplan, 1998). Because each interpersonal construct has a partic-

ular focus and is only useful to interpret certain aspects of social events (Adams-Webber, 2003; Kelly, 1955), individuals' capacity to understand diverse contexts depends on the number of different interpersonal constructs they have available (i.e., their interpersonal cognitive complexity; Crockett, 1965). Past research has, accordingly, shown interpersonal cognitive complexity to positively relate with individuals' ability to infer others' goals and perspectives from their observable conduct (e.g., Delia, Clark, & Switzer, 1979; Delia & Crockett, 1973; Hale & Delia, 1976; Woods, 1996). Moreover, such complexity enables individuals to adjust their behavior to diverse interaction contexts (Applegate, 1982; Applegate, Coyle, Seibert, & Church, 1989; Hale, 1980; Kline & Floyd, 1990) and to pursue complex communication strategies that can overcome differences in opinions and objectives (O'Keefe & Shepherd, 1987; Wilson, 1990).

Based on these arguments, one might conclude that interpersonal cognitive complexity will facilitate an individual's interteam coordination behavior. Cognitively complex individuals may be able to use a broad variety of interpersonal constructs to reconcile incompatible goals and expectations between members of different teams in order to coordinate distinct teams' actions. In combination with the ideas outlined in Hypothesis 1, interpersonal cognitive complexity may, therefore, function as a central mediating mechanism through which “individuals' past experiences in multiple functions . . . shape their understanding of the organizational environment” and enable them to “identify and approach” external members for interteam coordination purposes (Joshi et al., 2009: 746).

Notably, however, there are important theoretical reasons to anticipate that the relationship between interpersonal cognitive complexity and interteam coordination is not as straightforward as this argumentation would suggest. Researchers have noted that interpersonal cognitive complexity merely represents one's cognitive capacities and potentials, whereas personal construct theory does not specify how individuals will use these capacities (Burlinson & Denton, 1997; Kelly, 1955). In fact, Burlinson and Caplan (1998) explicitly argued that individuals may utilize their interpersonal cognitive complexity “for varied ends, and may even choose to not use it at all” (Burlinson & Caplan, 1998: 240). As such, the linkage between interpersonal cognitive complexity and specific social behaviors depends on factors that cannot be explained by personal construct theory alone (Burlinson & Denton, 1997;

Denton, Burleson, & Sprenkle, 1995). It appears necessary, therefore, to supplement predictions from personal construct theory with other theoretical perspectives that can explicate specific circumstances under which team members with broad functional experiences will be more or less motivated to direct their interpersonal cognitive complexity toward interteam coordination.

The Moderating Role of Organizational Identification

Extending personal construct theory: Incorporating a social identity perspective. In identifying potential moderating variables that may influence whether individual members direct their interpersonal cognitive complexity toward (or away from) interteam coordination, social identity theory (Tajfel, 1982; Turner et al., 1987) appears particularly useful. Personal construct theory explicates individuals' development of socio-cognitive capacities that can be used for social behavior in general and intergroup behavior in particular (Burleson & Caplan, 1998). Social identity theory shares this focus on social behavior (see Ashforth & Mael, 1989). In fact, social identity theory is among the most well established theoretical perspectives on intergroup relations (Richter, West, Van Dick, & Dawson, 2006), and it has been widely used to explain individuals' boundary spanning behavior (e.g., Drach-Zahavy, 2011; Fisher, Maltz, & Jaworski, 1997; Joshi et al., 2009; Joshi, 2006). Moreover, social identity theory incorporates key motivational factors known to direct individuals' social actions that are not covered by personal construct theory's mere cognitive focus (Ellemers, de Gilder, & Haslam, 2004). As such, a social identity approach can complement personal construct theory by uncovering key conditions under which individuals are more or less likely to utilize their cognitive potentials to benefit (or hamper) between-team coordination.

Specifically, social identity theory posits that people's identity, or their self-definition of who they are, depends on their membership of relevant social groups (Ashforth & Mael, 1989). The importance of a specific social group for an individual's identity is reflected in his or her identification with that group (Dutton, Dukerich, & Harquail, 1994). Further, such identification decisively influences individuals' behavior, with stronger identification motivating people to pursue beneficial outcomes for the group in question to ensure its success (i.e., the identity-matching principle; Ellemers & Rink, 2005;

Ullrich, Wieseke, Christ, Schulze, & Van Dick, 2007). An individual's strong identification with the overall organization, for example, is likely to motivate him or her to devote particular effort to the attainment of organizational goals (Christ, van Dick, Wagner, & Stellmacher, 2003; Riketta & Van Dick, 2005).

As noted before, it is a unique and defining feature of interteam coordination that such behavior aims at integrating distinct teams' actions to achieve collective, organizational outcomes (Ancona & Caldwell, 1992; DeChurch & Marks, 2006; Marks et al., 2005). Hence, although interteam coordination can have important, positive implications for a member's own team (Marrone et al., 2007), such activities are also critical for the organization as a whole (Marks et al., 2005; Mathieu et al., 2001). Based on the identity-matching principle, we therefore suggest that individual team members' *organizational* identification, in particular, will critically shape their willingness to engage in interteam coordination behavior (Ashforth, Harrison, & Corley, 2008; Ullrich et al., 2007). Specifically, we propose that organizational identification will motivate an individual to either direct his or her interpersonal cognitive complexity toward establishing effective coordination between teams (if organizational identification is high) or, conversely, toward avoiding such effortful behavior (if organizational identification is low).

The interactive relationship of cognitive complexity and organizational identification. On the one hand, we anticipate interpersonal cognitive complexity to positively relate with interteam coordination among team members with higher organizational identification. In this situation, a team member subsumes the organization into his or her self-image and experiences organizational successes and failures as personal successes and failures (Ashforth & Mael, 1989; Bartel, 2001). In order to maintain a positive identity, this person will be highly motivated to invest (cognitive) resources for organization-level outcomes (Ellemers & Rink, 2005; Ullrich et al., 2007). One primary way to realize such outcomes is by coordinating efforts with members from other teams in the organization (Marks et al., 2005; Mathieu et al., 2001; Smith et al., 1995).

By itself, however, a member's respective motivation may not reliably trigger interteam coordination. Such behavior is far from trivial, because distinct teams' members often hold incompatible perspectives and goals (Griffin & Hauser, 1996; Maltz & Kohli, 2000). Coordination between members of different teams may, thus, be troubled by

incongruent expectations and significant doubts about what and how to coordinate (Friedman & Podolny, 1992; Miles & Perreault, 1976). Members with high interpersonal cognitive complexity may enjoy a distinct benefit in dealing with these difficulties, utilizing the wide variety of interpersonal constructs at their disposal to effectively decode other teams' members' divergent opinions and behaviors and to manage complicated between-team situations (cf. Burleson & Caplan, 1998; O'Keefe & Sypher, 1981). Hence, cognitively complex team members who strongly identify with the overall organization may possess both the cognitive resources and the motivation required for interteam coordination. In contrast, team members with low interpersonal cognitive complexity may find it difficult to engage in interteam coordination behavior even if they are highly identified with the organization, given their limited capacity to make sense of complex and diverse social contexts. Logically, then, we would expect interteam coordination to be more pronounced among highly identified members with higher rather than lower interpersonal cognitive complexity.

On the other hand, we anticipate interpersonal cognitive complexity to negatively relate with interteam coordination among members with lower organizational identification. Such boundary spanning can add significantly to an individual's workload, thereby increasing stress and depleting valued resources (Levina & Vaast, 2005; Marrone et al., 2007; Singh, Goolsby, & Rhoads, 1994). Moreover, interteam coordination may compete for attention and energy with other, potentially more salient obligations (e.g., core job tasks; Faraj & Yan, 2009), leading individuals to perceive such behavior as threatening their own, personal success (Ramarajan, Bezrukova, Jehn, & Euwema, 2011). Team members with lower organizational identification attach little relevance to the attainment of organizational goals because their membership in the organization is largely irrelevant for their self-definition (Ashforth & Mael, 1989; Dutton et al., 1994; Ellemers et al., 2004). Consequently, for these individuals, the personal costs associated with interteam coordination may outweigh its organizational benefits (Fisher et al., 1997), motivating them to avoid such behavior as much as possible.

Importantly, however, we further propose that individual members' ability to circumvent interteam coordination hinges on their interpersonal cognitive complexity. In modern organizations with pronounced between-team interdependen-

cies, external demands to coordinate one's work activities with other teams are a common fact of life (Hoegl et al., 2004; Mathieu et al., 2001; Smith et al., 1995). Although such boundary spanning often does not belong to employees' main tasks, it is frequently expected by coworkers and organizational authorities (Ancona & Caldwell, 1988; Faraj & Yan, 2009). We anticipate that high cognitive complexity will help team members with low organizational identification to successfully resist these expectations. In particular, such members may deeply comprehend others' requests for cooperation, and, based on this understanding, conceive effective strategies to defy such demands (Kline, 1991; Kline & Chatani, 2001; O'Keefe & Delia, 1979). They may, for example, find it relatively easy to devise credible excuses or propose alternative courses of action that limit their own involvement in coordination activities. Empirical research provides some evidence for this "dark side" of interpersonal cognitive complexity, illustrating that cognitively complex individuals can use more effective refusal strategies to counter unwelcome requests (Kline & Floyd, 1990). Similarly, individuals have been shown to use their interpersonal cognitive complexity to discover others' weak spots and to stifle undesired interactions (Bacue & Samter, 2001; Burleson & Denton, 1997).

Individuals with less interpersonal cognitive complexity, in contrast, may lack the cognitive capacity to effectively guard themselves from interteam coordination demands. These persons may find it difficult to resist external coordination requests in an effective, socially sensible manner that prevents personal reprimands (e.g., loss of social support; Bouty, 2000). Even if low organizational identification motivates a team member to avoid interteam coordination, a lack of cognitive complexity may, thus, render it virtually impossible for the respective member to fully circumvent such behavior. All in all, we therefore expect interteam coordination to be less pronounced among individuals with lower organizational identification if they exhibit higher rather than lower interpersonal cognitive complexity.

Hypothesis 2. The relationship between an individual's interpersonal cognitive complexity and interteam coordination is moderated by his or her organizational identification. This relationship is positive when organizational identification is higher, but negative when organizational identification is lower.

A Moderated Mediation Model

Taken together, the present argumentation suggests that breadth of functional experience positively relates with an individual's interpersonal cognitive complexity (Hypothesis 1). Such complexity, in turn, is argued to positively associate with interteam coordination among members with higher organizational identification, but to negatively associate with such behavior among members with lower organizational identification (Hypothesis 2). Hence, interpersonal cognitive complexity may be an important mediating mechanism that transfers the role of broad functional experiences for interteam coordination—but the direction of this indirect linkage should be conditional on one's organizational identification.

Interestingly, the basic notion of a mediating role of cognitive complexity is consistent with the general theoretical assumption that demographic characteristics can influence an individual's behavior by shaping important cognitive processes (Hambrick & Mason, 1984; Markóczy, 1997), as well as with specific theoretical arguments that connect breadth of functional experience with interteam coordination (Griffin & Hauser, 1996; Joshi et al., 2009). Our argumentation suggests, however, that the role of broad functional experiences is more intricate than previous theorizing would suggest. It appears that breadth of functional experience, by promoting interpersonal cognitive complexity, may represent a mixed blessing, with the potential to either indirectly promote (with stronger organizational identification) or deteriorate (with lower organizational identification) an individual's interteam coordination.

Hypothesis 3. Organizational identification moderates the indirect relationship between an individual's breadth of functional experience and interteam coordination, as mediated by interpersonal cognitive complexity. This indirect relationship is positive when organizational identification is higher, but negative when organizational identification is lower.

Members' Overall Interteam Coordination and Team Performance

Although interteam coordination reflects individual behavior, members' respective actions can come to characterize a team as a whole and, thus, emerge as a bottom-up, team-level construct with distinct consequences for a team's outcomes (Joshi

et al., 2009). Hence, a complete understanding of interteam coordination requires an examination of both its individual-level antecedents and its team-level implications (Marrone et al., 2007). As noted earlier, interteam coordination aims at managing critical interdependencies between teams to achieve collective, organizational goals (Ancona & Caldwell, 1992; Marks et al., 2005; Mathieu et al., 2001). This organization-level focus notwithstanding, a team's own performance may also benefit if its members engage in interteam coordination behaviors to a greater extent.

In the process of managing organization-wide activities, for example, members' interteam coordination may provide opportunities for discussing the team's own tasks with external members and, subsequently, to resolve complexities and uncertainties in a team's work environment (Hoegl et al., 2004; Marks et al., 2005). Moreover, to the extent that a team's members, as a whole, engage in such boundary spanning, the team should gain improved access to sources of external support (e.g., assistance from other teams), enabling it to better execute crucial tasks (Joshi et al., 2009; Marrone, 2010). Members' overall level of interteam coordination may, therefore, help a team to achieve complex goals that it could not attain on its own. Consistent with this reasoning, empirical research at the team level of analysis has shown overall interteam coordination to positively associate with team performance across diverse organizational contexts (e.g., Hoegl et al., 2004; Marrone et al., 2007).

Hypothesis 4. A team's overall level of interteam coordination is positively associated with team performance.

STUDY 1: METHODS

Sample and Procedure

The first study examined Hypotheses 1–3 in the context of an international training exercise for peace support missions. This one-week exercise included an interagency coalition comprising 34 distinct teams. The training mission simulated the coalition's deployment into a fictitious country, with the main task of assisting an internationally recognized government in dealing with foreign aggression and restoring civil order. Major training goals were to optimize cooperation between military, government, and nongovernment agencies. As such, mission success was contingent on the successful integration of contributions from these diverse, in-

terdependent entities. The exercise's "comprehensive approach" philosophy therefore encouraged individual members to coordinate between teams, emphasizing the importance of interagency cooperation to achieve shared objectives (see Goodwin, Essens, & Smith, 2012). Participants were senior members of a military command-and-control center as well as several humanitarian aid foundations, police forces, local government associations, and the foreign affairs departments of multiple nations (i.e., the United States, Germany, and the Netherlands).

Targeted study participants were all 172 members of the exercise's coalition headquarters. Potential participants received a paper-and-pencil survey two days before the end of the exercise from an independent observer. We chose this timing of measurement because (a) critical mission tasks had been completed at this point, with ample potential for member cooperation, and (b) participants still had sufficient time to complete the survey before the end of the exercise. Participation was voluntary and anonymity assured. Sixty-nine participants, distributed across 22 teams, returned complete and usable surveys, for an effective response rate of 40%. The mean age of these participants was 43 years ($SD = 8.96$). On average, respondents had worked for about 17 years for their home agencies ($SD = 11.24$). Our sample included both civilian (36%) and military personnel (64%). Of the latter, 91% were officers (i.e., lieutenant or higher). Because less than 5% of the training exercise's participants were female, we were not permitted to ask for respondents' gender, to assure anonymity.²

Measures

Breadth of functional experience. Consistent with previous research (Bunderson, 2003; Bunderson & Sutcliffe, 2002), we measured breadth of functional experience by asking respondents to indicate their years of work experience in each of 11 functional domains (e.g., general management, per-

sonnel, logistics, planning, operations, finance, etc.). Subject matter experts had identified these domains as comprehensively covering all crucial functional aspects of the training exercise. On this basis, we calculated each person's breadth of functional experience, using Bunderson's (2003) version of Blau's (1977) heterogeneity index:

$$1 - \sum_{i=1}^k p_i^2$$

where p_i is the percentage of total years of work experience in the i th functional domain and k represents the total number of functional domains ($k = 11$ in the present case). This resulted in an overall score for an individual's breadth of functional experience on a scale ranging from 0 (i.e., all work experience gathered in a single functional domain) to a theoretical maximum of .91 (i.e., total work experience evenly distributed across all 11 domains).³

Interpersonal cognitive complexity. Consistent with common recommendations for the measurement of individuals' abilities (Mabe & West, 1982), we captured interpersonal cognitive complexity with Crockett's (1965) Role Category Questionnaire (RCQ), a performance-based test that prompts respondents to describe a liked and a disliked peer in as much detail as possible (Sypher & Sypher, 1988). The level of detail in participants' respective answers is subsequently used as an indicator of cognitive complexity (Burlinson & Denton, 1997). As recommended for research in organizational settings, we asked respondents to describe two existing persons (excluding themselves) they knew well from their current or previous work environment (Penley, Alexander, Jernigan, & Henwood, 1991; Sypher & Zorn, 1986; Zorn & Violanti, 1996). In the present study, for example, a participant described a liked colleague as "an analytical person; focused on others' interests; diplomatic" and a disliked colleague as "arrogant; unreliable; unable to keep focus." Research has shown that such descriptions contain representative samples of the interpersonal constructs a respondent has available to understand other individuals and social situations, thereby providing a good indication of a respondent's interpersonal cognitive complexity (Burlinson & Waltman, 1988; Crockett, 1965).

Demonstrating convergent validity, prior research has shown that RCQ scores are significantly

² To assess the extent to which our final sample was representative of the targeted study population, we attempted to collect demographic information for all training participants. Given the temporary, multinational, and multi-agency character of the exercise, however, such information was available for only 54 of the nonrespondents (52%). These 54 individuals did not differ significantly from our final sample in terms of age ($M = 43$ years, $SD = 9.74$), tenure ($M = 17$ years, $SD = 13.09$), and military rank (84% officers).

³ Maximum diversity is calculated as $(k - 1)/k = (11 - 1)/11 = .91$ (see Harrison & Klein, 2007).

related to other socio-cognitive abilities (e.g., social perspective taking; Hale & Delia, 1976; Kline, Pelias, & Delia, 1991). Furthermore, indicating discriminant validity, RCQ scores are not (or only weakly) associated with potential confounding factors, such as writing speed, verbal fluency, and general intelligence (Burleson & Waltman, 1988). Studies have also shown that the RCQ exhibits high test-retest reliability for short to medium time spans, although it is sensitive to changes in cognitive complexity that can accrue over longer-term periods (Duys & Hedstrom, 2000; O'Keefe, Shepherd, & Streeter, 1982). Consequently, the RCQ is generally regarded as superior to alternative measures (Burleson & Caplan, 1998; O'Keefe et al., 1982; Sypher & Sypher, 1988; Zhang, Xin, & Lin, 2012), and it has been used in "virtually all cognitive complexity research appearing in the human communication literature" (Burleson & Waltman, 1988: 1).

Given the current field study's context, we used an untimed version of the RCQ. Research has shown timed and untimed RCQ versions to produce almost identical results (Burleson & Waltman, 1988; O'Keefe et al., 1982). Following the procedure of Crockett, Press, Delia, and Kenny (1974), the first author and an independent rater (blind to the participants' scores on other study variables) coded the number of different interpersonal constructs (e.g., traits, behaviors, habits) in respondents' RCQ descriptions. Based on adequate interrater reliability ($ICC[2] = .96, p < .01$), we averaged both coders' ratings into a single measure of each respondent's interpersonal cognitive complexity. Further, as in previous work (e.g., Burleson & Waltman, 1988), we estimated internal consistency reliability by calculating the correlation between the number of interpersonal constructs respondents used to describe the liked and disliked person, respectively. This correlation was $.58 (p < .01)$, which is equivalent to a Cronbach's alpha of $.73$, closely matching reliability estimates in other RCQ studies (Burleson & Waltman, 1988).

Organizational identification. We captured organizational identification with Mael and Ashforth's (1992) six-item measure. Individuals were asked to report their identification with the training mission coalition (not their home agency) on a seven-point scale (1 = "completely disagree," 7 = "completely agree"). Subject matter experts confirmed that this coalition was highly relevant for participants and, thus, constituted a meaningful target for individuals' organizational identification; in fact, such iden-

tification was actively promoted by the coalition's top leadership through collective mission statements and goals, as well as shared symbols (e.g., mission badges) that emphasized the joint and unique character of the coalition. Example items were "When I talk about [name of coalition], I usually say 'we' rather than 'they,'" and "The successes of [name of coalition] are also my successes." Cronbach's alpha was $.84$.

Inter-team coordination. Theorists have argued that boundary-spanning behaviors, such as inter-team coordination, reflect the degree to which an individual maintains interactive linkages with external teams (Joshi et al., 2009; Marrone, 2010). Capturing external team linkages, correspondingly, represents one of the "predominant approaches" for measuring boundary spanning (Joshi et al., 2009: 753). Tsai (2002), for example, measured team members' knowledge sharing with external units through a network-type item that asked respondents to indicate all units with which they exchanged technology or know-how. Similarly, we used a network-type measure to assess individual members' task-related coordination linkages with other teams as an indication of their interteam coordination. We provided respondents with a list of the teams present during the exercise and asked them to select all teams with whose members they had regularly coordinated activities (see Wasserman & Faust, 1994). For greater clarity, we provided examples of relevant behaviors (e.g., aligning work activities, negotiating joint efforts, resolving joint problems; Ancona & Caldwell, 1992). An individual's degree of interteam coordination was calculated as the total number of external teams with which he or she had coordinated.

Respondents' interteam coordination scores were positively skewed (skewness = $1.09, p < .05$), violating assumptions of normality (Cohen, Cohen, West, & Aiken, 2003). We therefore applied a square root transformation on individuals' scores, which effectively reduced skewness (skewness = $0.10, ns$) and produced normally distributed regression residuals (see Cohen et al., 2003). We repeated all analyses using both the transformed and the untransformed scores, with largely equivalent results. Below, we report results based on the transformed interteam coordination measure; results using the untransformed measure are available from the first author.

We conducted a separate survey study within a military staff organization to validate this interteam coordination measure. In this study, we administered the respective instrument to 77 organizational

members. These respondents also completed the five-item task-coordination subscale of Ancona and Caldwell's (1992) boundary-spanning measure (sample item: "To what extent did you personally coordinate work activities with external teams"; 1 = "not at all" to 7 = "to a very large extent"; Cronbach's $\alpha = .93$). The interteam coordination measure used in our study correlated strongly with this more subjective scale ($r = .67$; $p < .01$), providing evidence for the convergent validity of the present measure (Nunnally, 1978).

Control variables. Resource dependency theory suggests that individuals may exhibit more interteam coordination behavior when they are dependent upon other teams (Joshi et al., 2009; McCann & Ferry, 1979). Interteam coordination may, thus, not be completely discretionary but may, at least partially, result from the interdependent nature of an individual's work. Therefore, we considered respondents' perceived interteam interdependence as a control variable. We adapted three items from van der Vegt, van de Vliert, and Oosterhof's (2003) task interdependence measure to capture this construct (1 = "completely disagree" to 7 = "completely agree"). Specifically, we used these statements: "In order to complete my work, I had to exchange information and advice with other groups," "I had to work closely with other groups to do my work properly," and "I rarely had to check or work with other groups" (reverse coded). Cronbach's α was .77.

Moreover, individuals with longer tenure in their parent agency and higher hierarchical positions may hold greater legitimacy to influence external team members (Ibarra, 1993; Spekman, 1979), potentially facilitating their interteam coordination (Ancona & Caldwell, 1992; Joshi et al., 2009; Levina & Vaast, 2005). Hence, we considered individuals' organizational tenure with their current employer (in years) and leadership position as control variables. To assess a participant's leadership position, individuals indicated the percentage of time they devoted to leadership activities during the exercise.

Data Analysis

Because individual respondents worked within teams, we calculated an intraclass correlation coefficient (i.e., ICC[1]) for our dependent variable to determine whether this nesting violated assumptions of independence. In line with previous research (e.g., Marrone et al., 2007), we found that individuals' interteam coordination behavior was

unaffected by their team membership ($\text{ICC}[1] = .00$, ns). As such, we used ordinary least squares regressions to test Hypotheses 1 and 2 (Cohen et al., 2003; Snijders & Bosker, 1999). Moreover, we employed Edwards and Lambert's (2007) moderated mediation procedure to test Hypothesis 3. We obtained bootstrapped parameter estimates for the indirect relationship between breadth of functional experience and interteam coordination (through interpersonal cognitive complexity) at both higher ($+1SD$) and lower ($-1SD$) levels of organizational identification, and we assessed the statistical significance of these estimates based on bias-corrected 95% confidence intervals. This method overcomes typical power problems caused by non-normal sampling distributions of indirect relations (Shrout & Bolger, 2002). We standardized all predictors prior to the analyses.

STUDY 1: RESULTS

Descriptive Statistics

Means, standard deviations, and bivariate correlations for all Study 1 variables are reported in Table 1. As expected, breadth of functional experience was positively associated with interpersonal cognitive complexity ($r = .27$, $p < .05$). Of the control variables, only perceived interteam interdependence exhibited a significant relation with an outcome variable in our model (i.e., cognitive complexity; $r = .27$, $p < .05$). We therefore retained perceived interteam interdependence in hypotheses testing and excluded the other, nonsignificant covariates to avoid biased parameter estimates (Becker, 2005). Notably, results remained virtually unchanged when also excluding perceived interteam interdependence or incorporating all controls.

Hypotheses Testing

Hypothesis 1 posited that an individual team member's breadth of functional experience is positively related to his or her interpersonal cognitive complexity. As shown in Table 2, breadth of functional experience and interpersonal cognitive complexity were indeed positively associated ($B = .39$, $SE = .16$, $p < .05$), thus supporting our first hypothesis.

Hypothesis 2 predicted organizational identification to moderate the relationship between interpersonal cognitive complexity and interteam coordination. As depicted in Table 2, the interaction

TABLE 1
Means, Standard Deviations, and Correlations (Individual-level Variables, Studies 1 and 2)

Variable	Study 1 <i>M (SD)</i>	Study 2 <i>M (SD)</i>	<i>r</i>							
			1	2	3	4	5	6	7	8
1. Leadership position	.29 (.36)	.21 (.41)		.30*	-.14	.25*	-.01	.25*	.16	
2. Organizational tenure	17.22 (11.24)	12.90 (10.92)	.01		-.31*	.20	-.14	.05	.07	
3. Perceived interteam interdependence	5.43 (1.41)	4.72 (1.56)	.04	-.08		-.04	.27*	.26*	.01	
4. Breadth of functional experience	.59 (.18)	.27 (.27)	.11	-.12	.10		.27*	.17	.17	
5. Cognitive complexity	3.83 (1.38)	5.05 (2.32)	.02	-.15	.05	.29**		.10	.27*	
6. Organizational identification	4.36 (1.13)	4.93 (1.14)	.23*	.18*	.12	.22*	.13		.23	
7. Interteam coordination	3.47 (1.21)	2.82 (1.58)	.18*	.11	.23*	.14	.07	.20*		
8. Gender	—	.42 (.50)	-.20*	-.08	-.02	-.13	.12	-.05	-.16	
9. Education level	—	2.67 (.61)	.02	-.18*	.07	.03	.16	.17	.10	-.14

Note: Study 1: Correlations shown above the diagonal; $n = 64$ – 69 . Study 2: Correlations shown below the diagonal; $n = 120$ – 121 .

* $p < .05$

** $p < .01$

coefficient for cognitive complexity $\times$ organizational identification was significantly associated with interteam coordination ($B = .45$, $SE = .14$, $p < .01$). As expected, Figure 2 (upper half) illustrates that interpersonal cognitive complexity was positively related to interteam coordination when organizational identification was higher (simple slope at $+1SD$: $B = .70$, $SE = .19$, $p < .01$; see Aiken & West, 1991). Contrary to our expectations, this relationship was nonsignificant (rather than significantly negative) when identification was lower (simple slope at $-1SD$: $B = -.20$, $SE = .23$, ns). Hence, these results partially supported Hypothesis 2.

Finally, Hypothesis 3 suggested a conditional indirect relationship, whereby the indirect association between breadth of functional experience and interteam coordination, through cognitive complexity, is contingent on the value of organizational identification. As predicted, we found a significant and positive indirect relationship when organizational identification was higher (indirect relationship at $+1SD = .27$; 95% confidence interval = .04 to .65). The indirect relationship was not significantly different from zero (rather than significantly negative), however, when organizational identification was lower (indirect relationship at $-1SD = -.08$; 95%

TABLE 2
Regression Results and Conditional Indirect Relationship (Study 1)

Predictor	Cognitive Complexity	Interteam Coordination	
Perceived interteam interdependence	.41 (.16)*	-.15 (.16)	-.14 (.15)
Breadth of functional experience	.39 (.16)*	.06 (.15)	.09 (.15)
Cognitive complexity (CC)		.33 (.16)*	.25 (.15)
Organizational identification (OI)		.28 (.15)	.42 (.15)**
CC $\times$ OI			.45 (.14)**
Conditional indirect relationship at OI $\pm 1SD$.			

	95% Boot Confidence Interval ^a		
	Boot Indirect Relation	Lower Bound	Upper Bound
$-1SD (-1.00)$	-.08	-.36	.06
$+1SD (+1.00)$	.27	.04	.65

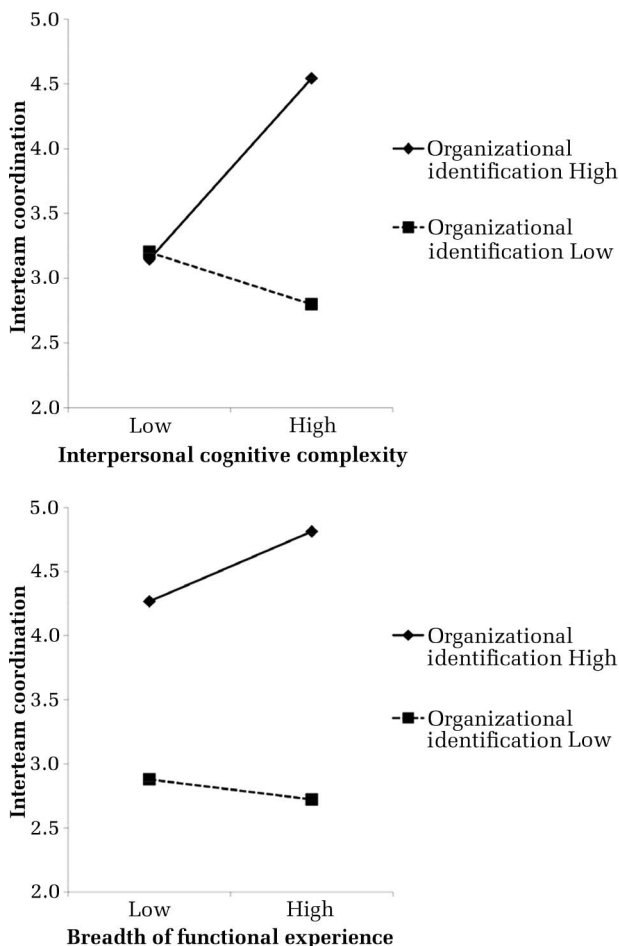
Note: $N = 67$ individuals. Unstandardized regression coefficients are shown; standard errors are noted within parentheses.

^a Based on 1,000 bootstrap samples.

* $p < .05$

** $p < .01$

FIGURE 2
Breadth of Functional Experience, Interpersonal Cognitive Complexity, Organizational Identification, and Interteam Coordination (Study 1)



confidence interval = $-.36$ to $.06$). Thus, Hypothesis 3 received partial support (see Figure 2, lower half).

STUDY 2: PURPOSE AND METHODS

Purpose of Study 2

We note that Study 1 was constrained to a training mission context with limited duration and employed a relatively small sample. Also, the first study's research setting required a somewhat crude measure of interteam coordination, asking participants to broadly identify the external teams (rather than specific members) with which they had coordinated. Study 2 aimed to address these limitations, corroborating the generalizability and robustness of

Study 1's findings within a more traditional organizational context, using a larger sample and a finer-grained interteam coordination measure. Moreover, we were able to gather team performance information in Study 2. Hence, beyond testing Hypotheses 1–3 at the individual level of analysis, this second study enabled us to examine the team-level performance consequences of members' overall interteam coordination efforts postulated in Hypothesis 4.

Sample and Procedure

The second study took place in a Dutch municipality administration serving a city with approximately 80,000 citizens. Main tasks of the administration included preparing and executing city council decisions, implementing national legislation, and providing a wide array of public services. Similar to many public administrations around the world, the complex nature of today's societal problems, in combination with budgetary cutbacks, had led our host organization to make drastic changes in its operations. The organization aimed to simultaneously achieve cost savings and higher effectiveness by using a team-based structure, joint working arrangements, and other forms of cross-disciplinary, boundary-spanning collaborations (cf. McGuire, 2006). Consequently, demands for interteam coordination were an integral part of employees' daily work. The municipality administration's top management agreed to this study in return for feedback on their efforts to improve intra-organizational collaboration.

Because management limited the number of individuals we could include in our study, we targeted 295 randomly selected office employees across the organization. Participation was voluntary and anonymity assured. Of those employees, 121 (distributed over 36 of the organization's 40 teams) provided complete responses, resulting in an effective response rate of 41%. Respondents were, on average, 48 years of age ($SD = 10.33$), had worked with the municipality administration for an average of about 13 years ($SD = 10.92$), and were mostly male (58%). Company records confirmed that respondents did not differ significantly from nonrespondents in terms of age (nonrespondents' mean age = 47 years, $SD = 9.84$), gender (51% male), and organizational tenure ($M = 14$ years, $SD = 11.58$).

Measures

Breadth of functional experience. Similar to Study 1, we measured breadth of functional experience by asking respondents to report their years of work experience in different functional domains. We focused on 16 functional domains identified by organizational representatives as particularly relevant for the organization (e.g., general management, finance, administrative support, legal, personnel, public safety, etc.). On this basis, we calculated Bunderson's (2003) heterogeneity index to capture an individual's breadth of functional experience.

Interpersonal cognitive complexity. We employed the same methodology as in Study 1 to assess interpersonal cognitive complexity (Crockett, 1965). Again, the first author and an independent rater coded the number of distinct interpersonal constructs in respondents' RCQ answers. With adequate inter-rater reliability ($ICC[2] = .97$, $p < .01$), these two ratings were averaged into a single score for each participant. To gauge internal consistency reliability, we again calculated the correlation between the number of interpersonal constructs used when describing a liked and a disliked coworker, respectively ($r = .74$, $p < .01$; equivalent to a Cronbach's alpha of .85).

Organizational identification. We measured organizational identification with Mael and Ashforth's (1992) six-item scale, as in Study 1. The item "If a story in the media criticized the organization, I would feel embarrassed" was excluded because organizational representatives had reservations. Cronbach's alpha was .87.

Interteam coordination. We used a similar network-type measure as in Study 1 to quantify individuals' interteam coordination behavior. In the present study, however, we asked respondents to identify the specific *individuals* from other teams with whom they regularly coordinated when "preparing policy advice for management or the city council, completing joint projects, participating in task groups, and/or executing day-to-day tasks" (see Hansen, Mors, & Løvås, 2005, for a similar approach). During pre-survey interviews with 14 employees (conducted 4 weeks before the survey), interview partners had indicated that work-related coordination between teams primarily occurred during these activities. We calculated interteam coordination as the total number of external contacts selected by each respondent.

As in Study 1, individuals' interteam coordination scores were positively skewed (skewness =

2.14, $p < .05$). Using the procedure outlined in Study 1, we therefore again computed the square root of individuals' respective scores. This transformation reduced skewness (skewness = 0.40, *ns*) and resulted in normally distributed regression residuals (Cohen et al., 2003). We repeated our analyses using both transformed and untransformed interteam coordination scores, with largely equivalent results. Results using transformed scores are reported in the following; untransformed results are available from the first author.

To assess comparability between the measurement approaches used in Studies 1 and 2, we calculated the total number of teams (rather than individuals) with which respondents had coordinated as an alternative measure of interteam coordination, as outlined in Study 1. We found a strong correlation between these two measures in the Study 2 dataset ($r = .86$, $p < .01$). Furthermore, we ran Study 2's analyses using both the individual-based and the team-based measure, which led to the same conclusions. Due to its greater precision, we report results using the individual-based measure in the following.

Finally, to examine Hypothesis 4, we derived a *team-level* measure of overall interteam coordination (i.e., the degree to which a team's members, as a whole, engaged in such behavior). Building on research on inter-unit knowledge sharing (Hansen, 2002; Tsai, 2002), we operationalized overall interteam coordination as the total number of teams with which the focal team had coordinated. Theorists typically cast boundary spanners as acting on behalf of their team as a whole (Marrone, 2010). Hence, we counted a between-team coordination linkage if at least one member of the focal team indicated coordinating with another team, or if at least one member of the other team indicated coordinating with the focal team. Besides the 121 respondents for the individual-level analyses reported previously, an additional 84 employees had provided information on their interteam coordination behavior (although they did not complete other focal measures, such as the relatively labor-intensive cognitive complexity test). To obtain a more complete picture, we included these additional respondents in the team-level analyses.⁴

Team performance. We obtained data on teams' performance from the host organization's official

⁴ The additional respondents did not differ significantly from the initial sample in terms of gender, age, tenure, and interteam coordination behavior.

performance appraisal system. Team performance was captured in this system as members' average team-oriented performance (i.e., a compilation of individual members' effectiveness on tasks directed at team goals; DeShon, Kozlowski, Schmidt, Milner, & Wiechmann, 2004; Klein & Kozlowski, 2000). Specifically, team performance scores reflected (a) members' average contribution to collective, team-level outcomes and (b) their average effectiveness at realizing goals specified in a team's strategic plan (rated by supervisors on a scale from 1 = "insufficient" to 4 = "excellent"). Management used teams' average scores on these two items to monitor overall team performance and provide team feedback. We received team performance scores for 33 of our sample's teams (gathered approximately three months after our initial survey), as well as anonymized ratings of the 454 individuals on which these team scores were based ($M = 14$ ratings per team, $SD = 9.39$; $ICC[1] = .19$, $p < .01$, $ICC[2] = .77$, median $r_{wg(j)} = .87$). Because the two items' standard deviations differed strongly, we standardized both items before averaging them (Anderson & Dill, 2000; Cortina, 1993). The items were positively correlated ($r = .43$, $p < .05$). This parallels a Cronbach's alpha of .61, comparable to other studies using organizational performance appraisals (e.g., Parker & Axtell, 2001).

Control variables. For the individual-level analyses (Hypotheses 1–3), we measured respondents' perceived interteam interdependence (Cronbach's alpha = .83) and organizational tenure as potential control variables, following the procedures outlined in Study 1. In addition, we gathered archival information about respondents' formal leadership positions (0 = not in a leadership position; 1 = in a leadership position), gender (0 = male; 1 = female), and education (1 = lower education; 2 =

middle education; 3 = higher education). For the team-level analyses (Hypothesis 4), we gathered teams' size from company records and considered it as a control variable because size may determine a team's resource base (Marrone et al., 2007). Further, because members' interdependence and tenure may shape a team's coordination efforts and performance (Joshi et al., 2009), we calculated members' average interteam interdependence ($ICC[1] = .10$, $p < .05$, $ICC[2] = .39$, median $r_{wg(j)} = .71$) and average organizational tenure as possible team-level covariates.

STUDY 2: RESULTS

Descriptive Statistics

Table 1 presents means, standard deviations, and bivariate correlations for the individual-level variables in Study 2. As expected, breadth of functional experience was positively related to interpersonal cognitive complexity ($r = .29$, $p < .01$). Of the control variables, perceived interteam interdependence ($r = .23$, $p < .05$) and leadership position ($r = .18$, $p < .05$) were significantly related to individuals' interteam coordination. Gender, education, and organizational tenure were unrelated to Study 2's individual-level outcome variables, and, therefore, excluded from further analyses (see Becker, 2005). Bivariate team-level correlations revealed that none of the control variables were significantly related to team performance (see Table 3). Hence, the team-level controls were omitted from further analyses (Becker, 2005).

Hypotheses Testing

Mirroring Study 1, individuals' interteam coordination behavior was unaffected by their team

TABLE 3
Means, Standard Deviations, and Correlations (Team-level Variables, Study 2)

Variables	<i>M</i>	<i>SD</i>	<i>r</i>			
			1	2	3	4
1 Team size	19.05	10.88				
2 Average organizational tenure	13.10	6.73	.05			
3 Average interteam interdependence	4.57	1.07	-.19	.04		
4 Overall interteam coordination	21.10	8.41	.24	.02	.44**	
5 Team performance	.00	.85	-.03	.04	.05	.44*

Note: $n = 33$ to 40 teams due to missing values.

* $p < .05$

** $p < .01$

membership ($ICC[1] = .00, ns$). Hence, we used the same regression procedures as in Study 1 to test Hypotheses 1–3. As shown in Table 4, results revealed a positive relationship between an individual's breadth of functional experience and interpersonal cognitive complexity ($B = .67, SE = .21, p < .01$). Thus, Hypothesis 1 was supported.

As further shown in Table 4, we found an interactive relationship of cognitive complexity and organizational identification with an individual's interteam coordination behavior ($B = .53, SE = .15, p < .01$). As illustrated in Figure 3 (upper part), interpersonal cognitive complexity was positively related to an individual's interteam coordination when organizational identification was higher (simple slope at $+1SD$: $B = .49, SE = .19, p = .01$) but negatively related to interteam coordination when organizational identification was lower (simple slope at $-1SD$: $B = -.57, SE = .22, p = .01$). Therefore, we found supporting evidence for Hypothesis 2.

As depicted in the lower part of Table 4, tests of Hypothesis 3 showed that the conditional indirect relationship of breadth of functional experience with individuals' interteam coordination behavior, through interpersonal cognitive complexity, was positive when an individual's organizational identification was higher (indirect relationship at $+1SD = .33$; 95% confidence interval = .06 to .86). Furthermore, this conditional indirect relationship was negative when organizational identification was lower (indirect relationship at $-1SD = -.38$; 95% confidence interval = -1.00 to -.09).

This pattern is depicted in Figure 3 (lower part); it is fully in line with Hypothesis 3. We note that the results for Hypotheses 1–3 remained virtually unchanged when excluding or including all individual-level controls.

Finally, Hypothesis 4 predicted that a team's overall interteam coordination would positively relate to team performance. As shown in Table 3, we observed a positive team-level correlation between these constructs ($r = .44, p < .05$). Given that none of the team-level covariates were significantly related with team performance, this finding supports Hypothesis 4. Parenthetically, a team-level regression that included all team-level covariates also yielded a positive association between overall interteam coordination and team performance ($B = .47, SE = .16, p < .01$).

GENERAL DISCUSSION

Theoretical Contributions

The present research offers new perspectives on the individual-level origins of interteam coordination. As such, it extends prior work that has typically focused on boundary spanning as a collective, team-level phenomenon (e.g., Ancona & Caldwell, 1992; Drach-Zahavy, 2011; Hoegl et al., 2004; see Marrone et al., 2007, for an exception). These previous studies have provided important insights,

TABLE 4
Regression Results and Conditional Indirect Relationship (Study 2)

Predictor	Cognitive Complexity	Interteam Coordination	
Leadership position	-.03 (.21)	.22 (.14)	.21 (.14)
Perceived interteam interdependence	.05 (.21)	.33 (.14)*	.32 (.13)*
Breadth of functional experience	.67 (.21)**	.11 (.15)	.10 (.14)
Cognitive complexity (CC)		.04 (.15)	-.04 (.14)
Organizational identification (OI)		.20 (.15)	.27 (.14)
CC $\times$ OI			.53 (.15)**
Conditional indirect relationship at $OI \pm 1 SD$.			
95% Boot Confidence Interval ^a			
	Boot Indirect Relation	Lower Bound	Upper Bound
-1SD (-1.00)	-.38	-1.00	-.09
+1SD (+1.00)	.33	.06	.86

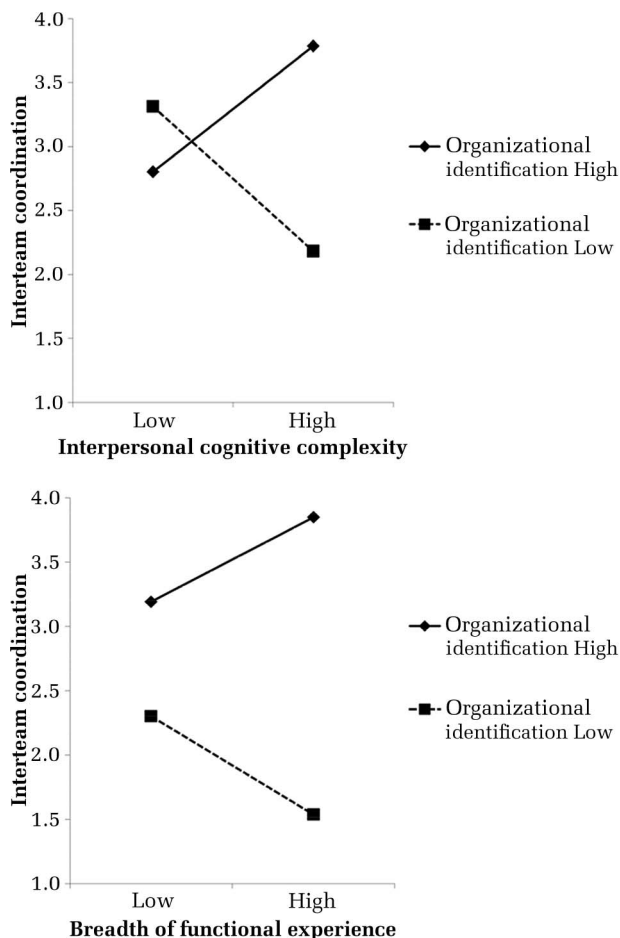
Note: $N = 121$ individuals. Unstandardized regression coefficients are shown; standard errors are noted within parentheses.

^a Based on 1,000 bootstrap samples.

* $p < .05$

** $p < .01$

FIGURE 3
Breadth of Functional Experience, Interpersonal Cognitive Complexity, Organizational Identification, and Interteam Coordination (Study 2)



and our team-level findings on the relationship between members' overall interteam coordination and team performance in Study 2 underline the relevance of such work. Research on the team-level antecedents of boundary spanning, however, cannot account for the fact that a team's respective activities are inherently an emergent, bottom-up phenomenon that originates from members' individual behaviors (Joshi et al., 2009; Marrone, 2010). Hence, little remains known about the "pivotal role of individual actors" in the development of a team's boundary-spanning activities (Williams, 2002: 103; see also Marrone, 2004). Both of the studies in the present investigation address this issue by highlighting individual members' breadth of functional experience, interpersonal cognitive

complexity, and organizational identification as key antecedents of their interteam coordination behavior, explicating why some members may contribute more or less to their team's overall boundary-spanning efforts than others. The current paper therefore adds to existing research on team boundary spanning by uncovering crucial, heretofore unexplored individual-level foundations of such activities.

These results may have important implications for the way scholars think about the role of broad functional experiences for individuals' interteam coordination behavior. Although breadth of functional experience is often assumed to promote interteam coordination (Burke & Steensma, 1998; Raskas & Brick, 1992), other authors have emphasized crucial disadvantages in this regard (Buyl et al., 2011), and empirical studies in related areas point toward considerable inconsistency (e.g., Leenders & Wierenga, 2002; Maltz & Kohli, 2000; Parry & Song, 1993). In an attempt to resolve this ambiguity, the present investigation indicates that the role of broad functional experiences may be more complex than previous theorizing would suggest (e.g., Griffin & Hauser, 1996; Joshi et al., 2009). On the one hand, our studies show that the breadth of functional experience–interteam coordination linkage is an indirect one and is driven by an individual's interpersonal cognitive complexity as a key mediator. On the other hand, this cognitive mechanism does not appear sufficient to reliably translate an individual team member's breadth of functional experience into interteam coordination. Individuals may only utilize the cognitive complexity arising from their broad experiences for such behavior if they strongly identify with the organization as a whole and, thus, are motivated to actively direct their cognitive capacity toward the overall organization's benefits.

Moreover, as shown in Study 2 (and indicated by a nonsignificant trend in Study 1), the positive role of interpersonal cognitive complexity for interteam coordination may reverse among team members with lower organizational identification. Our investigation therefore points toward a potential dark side of broad functional experiences and the associated cognitive complexity that previous research has overlooked. Team members whose identification with the organization is low may view interteam coordination in a pronouncedly negative light, focusing mainly on related efforts and drawbacks (Marrone et al., 2007; Singh et al., 1994). Superior cognitive complexity may enable these

individuals to shield themselves from external coordination demands—at the potential cost to both their own team and the organization as a whole (Marks et al., 2005; Smith et al., 1995). We acknowledge that this pattern of relationships should be considered tentative, given our two studies' somewhat inconsistent findings in this regard. One may speculate that this difference is attributable to Study 1's smaller sample size and lower statistical power. Also, the context of Study 1 may have imposed greater demands for interteam coordination, given that between-team collaboration was an explicit training goal. As such, it seems both theoretically and practically important to further examine the possibility that broad functional experiences may not be unequivocally beneficial, and that, in fact, the associated cognitive capacity may prevent rather than promote some members' boundary spanning.

Finally, our findings provide new insights into the origins of interpersonal cognitive complexity. Previous research has predominantly examined this construct as a determinant of individuals' communicative functioning (Burleson, 2007; Zhang et al., 2012). The few studies investigating predictor variables have typically examined individuals' early development during childhood and adolescence (Scarlett, Press, & Crockett, 1971; Turner, 2008; but see Goodwin & Ziegler, 1998), although theorists have emphasized that individuals' cognitive complexity may advance even during later life stages (Duys & Hedstrom, 2000; Fournier, 1996). To date, we know very little about key factors that may promote such development and, in turn, help individuals to achieve work-related outcomes (Sypher & Zorn, 1986). Burleson (2007: 124) has bemoaned this situation as “embarrassing and unacceptable,” arguing that “much more research is needed to better understand the antecedents of interpersonal cognitive complexity.” The present investigation takes steps toward addressing this issue, demonstrating a positive association between working adults' breadth of functional experience and interpersonal cognitive complexity.

Beyond the specific relationships uncovered, this investigation makes relevant contributions to broader theory development. Despite its popularity in other research areas (Walker & Winter, 2007), personal construct theory has rarely been used to examine organizational phenomena (see Dunn & Ginsberg, 1986, for an exception). Our findings show that this approach provides a useful starting point for understanding the mechanisms that link

employees' demographic characteristics with their behaviors in organizations. At the same time, our studies demonstrate that important conceptual extensions are needed to fully utilize this theory's explanatory power. In interteam settings, integrating personal construct theory with insights from social identity theory appears particularly fruitful. Personal construct theory specifies the cognitive underpinnings of interteam coordination, but it neglects motivational factors that may direct individuals' cognitive potentials toward such behavior. Social identity theory, in contrast, specifies why members may be motivated for (or against) interteam coordination, but it disregards individuals' cognitive capacities. Hence, combining these theories leverages their unique strengths while overcoming their distinct limitations, enabling a more complete depiction of the origins of individuals' boundary spanning behaviors. Taken together, we hope the conceptual advances in our studies will encourage further, constructive applications of personal construct theory in other areas of organizational research.

Limitations

Although this research has several methodological strengths (e.g., constructive replication of the results for Hypotheses 1–3 across two independent samples), some limitations should be noted. The design of the studies precludes causal conclusions, with all individual-level variables collected at one point in time. Notably, our model is in line with previous theoretical considerations that cast breadth of functional experience as an antecedent of boundary spanning (Joshi et al., 2009) and with general theoretical frameworks that position demographic characteristics as antecedents of individuals' cognition and behavior (Hambrick & Mason, 1984; Markóczy, 1997). Also, team performance was measured after the other variables in Study 2, rendering reverse causality less likely with regard to Hypothesis 4. We therefore have some confidence in the proposed direction of linkages. Nevertheless, we recognize the need for longitudinal or experimental evidence before causal inference is warranted.

Our studies' relatively low response rates may be another source of concern. Importantly, we note that potential participants for Study 1 all held high managerial ranks (e.g., diplomats and military officers), making it notoriously difficult to obtain high response rates (Cycyota & Harrison, 2006). Furthermore, the relatively time-consuming nature of the

interpersonal cognitive complexity test (i.e., a free-response format) accounts for the low response rates to some extent. Respondents' and nonrespondents' demographic characteristics were highly similar across both studies, ameliorating potential concerns about response bias (Armstrong & Overton, 1977).

Further, the fact that all individual-level data were gathered from the same employees might raise concerns about common source bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Both the independent variable (breadth of functional experience) and the outcome variable (interteam coordination), however, reflected direct accounts of factual information, not subjective perceptions or attitudes. Moreover, we used a performance-based (rather than self-reported) measure of interpersonal cognitive complexity. As such, common source variance is unlikely to bias study findings (Spector, 2006). In addition, common source variance is highly unlikely to account for the moderated relationships reported in this paper (see Evans, 1985), and, in Study 2, we used supervisors' independent ratings of team performance as an additional outcome variable. Nevertheless, future research might benefit from corroborating the present results using alternative measures (e.g., peer-ratings of boundary-spanning behavior).

Another possible limitation is that Hypothesis 4 (on the team-level linkage between overall interteam coordination and team performance) was only tested in Study 2. Also, the measure of team performance used for this analysis was developed by the host organization for applied purposes, and the team-level sample size was relatively small. This measure had, however, high practical relevance as an important element in the organization's performance management system (Parker & Axtell, 2001). Moreover, the relationship we uncovered is consistent with previous research (e.g., Ancona & Caldwell, 1992; Hoegl et al., 2004), supporting the validity of the present findings.

Finally, although key parts of the research model were tested across two diverse field settings, our findings may not translate to all organizational contexts. For example, teams in both of our samples were relatively stable and comprised members with long organizational tenure. Additional research is therefore needed to scrutinize our model in contexts with flexible and dynamic team memberships (see Matusik & Hill, 1998; O'Leary, Mortensen, & Woolley, 2011) before generalization to such settings is warranted.

Future Research Directions

Beyond addressing these limitations, future research could build on our findings in various ways. First, it seems particularly worthwhile to further investigate the negative indirect relationship between breadth of functional experience and interteam coordination observed among low identifiers in Study 2. Scholars could, for example, examine the specific boundary-management tactics that individuals use to counter interteam coordination requests (e.g., specific refusal strategies; Ancona & Caldwell, 1988; Faraj & Yan, 2009; Kline & Floyd, 1990). Such research may help to clarify not only *why* some individuals avoid boundary-spanning responsibilities but also *how* they achieve this goal. Subsequent research could also examine contextual factors that may influence the dark side of broad functional experiences and interpersonal cognitive complexity. Employees may, for example, be more prone to use such complexity to avoid interteam coordination when they perceive other organizational members as primarily acting politically and maximizing their self-interest (see Parker, Dipboye, & Jackson, 1995). Conversely, cognitively complex employees may have less opportunity to avoid interteam coordination when such behavior is strictly mandated; for example, by organizational or team policies enforcing and/or rewarding the respective acts.

Moreover, future research could expand the present model beyond interteam coordination. In particular, broad functional experiences and interpersonal cognitive complexity might also bear relevance for ambassadorial and scouting behaviors, given that such boundary-spanning activities also require productive interactions with external others (Ancona & Caldwell, 1988; Marrone, 2010). Relatedly, scholars could investigate additional contingency factors that may shape team members' use of broad functional experiences (and cognitive complexity) for different types of boundary spanning. As noted before, members may pursue organization-wide benefits with interteam coordination, but they may engage in scouting and ambassadorial activities primarily to enhance their own team's resources. Based on social identity theory's matching principle, members' willingness to pursue the latter types of behavior may therefore depend on their attitudes toward their own team rather than to the organization (Ullrich et al., 2007). Future research examining these possibilities could facilitate a more comprehensive view of the relevance of broad functional experiences in boundary-spanning contexts.

In addition, it may be interesting to extend our conceptual model by investigating contextual boundary conditions for the team-level relationship between overall interteam coordination and team performance. Prior theoretical work suggests, for example, that interteam coordination may be particularly valuable when a team faces complex task environments with multiple, unpredictable, and demanding stakeholders (Choi, 2002). Empirical research that builds on these notions may provide important, new insights on when a team's interteam coordination is more or less important for its performance outcomes (see Joshi et al., 2009; Marone, 2010).

Finally, it is worth mentioning that the measurement of interpersonal cognitive complexity may be challenging in organizational research. The RCQ (Crockett, 1965) employed in the present studies, in particular, is relatively labor intensive for participants due to its open-response format; in fact, our limited response rates are partially due to missing data on this instrument. On the other hand, the RCQ is generally seen as exhibiting higher construct validity and flexibility (Burleson & Waltman, 1988; Zhang et al., 2012), as compared with alternative interpersonal cognitive complexity measures (e.g., the Role Construct Repertory Test; Bieri, 1955; Goodwin & Ziegler, 1998). Hence, it appears worthwhile to seek for creative ways of resolving the potential difficulties surrounding the RCQ's usage in organizational settings, such as establishing high-quality, trusting relationships with participants, demonstrating personal commitment in the research process, and carefully delimiting a survey's overall length.

Practical Implications

This study provides practical implications for the selection, training, and management of organizational members required to engage in interteam coordination (e.g., members of project teams or multiteam systems; Davison, Hollenbeck, Barnes, Slesman, & Ilgen, 2012; Hoegl et al., 2004). Selecting individuals with broad functional experiences and advanced interpersonal cognitive complexity for such positions may be a first step toward facilitating such behavior. Also, organizations may draw on human resource development instruments (e.g., cross-functional job rotation, personnel movement, and training programs) to socialize employees within various functional areas. These programs may provide employees with the diverse social ex-

periences needed to advance their interpersonal cognitive complexity and interteam coordination capacity.

Our findings further illustrate, however, that such selection and training procedures may not suffice to promote interteam coordination. Employees with broad functional experiences and high cognitive complexity are less likely to engage in such behavior—and may even take active attempts to withdraw from interteam coordination—if they do not identify with the overall organization. Hence, organizations are well advised to work toward strengthening employees' organizational identification. Managers, as key representatives of the organization, play an important role in this regard. For example, managers could explicitly stress the distinctiveness and prestige of the organization toward their subordinates (Ashforth & Mael, 1989; Mael & Ashforth, 1992), emphasize teams' mutual goals, and use terms such as “we” instead of “us and them” when referring to multiple teams within the organization (Gaertner, Dovidio, Banker, Houlette, Johnson, & McGlynn, 2000).

CONCLUSION

The present research has examined mediating and moderating mechanisms in the relationship between an individual team member's breadth of functional experience and interteam coordination, and it has investigated team-level performance implications of such behavior. In doing so, our studies contribute to an integrative understanding of the role of broad functional experiences and underline the importance of interteam coordination. With higher organizational identification, broad functional experiences may indirectly and positively relate with interteam coordination behavior by shaping an individual's interpersonal cognitive complexity. In contrast, a negative indirect relationship may emerge between breadth of functional experience and interteam coordination under conditions of lower organizational identification. We hope these findings will stimulate further research on employees' boundary-spanning efforts and help organizations and managers to facilitate and benefit from the respective behavior.

REFERENCES

- Adams-Webber, J. R. 2003. Research in personal construct psychology. In F. Fransella (Ed.), *Interna-*

- tional handbook of personal construct psychology:** 51–58. Sussex, UK: John Wiley & Sons, Ltd.
- Aiken, L. S., & West, S. G. 1991. **Multiple regression: Testing and interpreting interactions.** Newbury Park, CA: Sage Publications.
- Ancona, D. G., & Caldwell, D. F. 1988. Beyond task and maintenance: Defining external functions in groups. **Group and Organization Management**, 13: 468–494.
- Ancona, D. G., & Caldwell, D. F. 1992. Bridging the boundary: External activity and performance in organizational teams. **Administrative Science Quarterly**, 37: 634–665.
- Anderson, C. A., & Dill, K. E. 2000. Video games and aggressive thoughts, feelings, and behavior in the laboratory and in life. **Journal of Personality and Social Psychology**, 78: 772–790.
- Applegate, J. L. 1982. The impact of construct system development on communication and impression formation in persuasive contexts. **Communication Monographs**, 49: 277–289.
- Applegate, J. L., Coyle, K., Seibert, J. H., & Church, S. M. 1989. Interpersonal constructs and communicative ability in a police environment: A preliminary investigation. **International Journal of Personal Construct Psychology**, 2: 385–399.
- Armstrong, J. S., & Overton, T. S. 1977. Estimating non-response bias in mail surveys. **Journal of Marketing Research**, 14: 396–402.
- Ashforth, B. E., Harrison, S. H., & Corley, K. G. 2008. Identification in organizations: An examination of four fundamental questions. **Journal of Management**, 34: 325–374.
- Ashforth, B. E., & Mael, F. 1989. Social identity theory and the organization. **Academy of Management Review**, 14: 20–39.
- Bacue, A., & Samter, W. 2001. **The dark side of cognitive complexity, II: The production of guilt-inducing messages.** Paper presented at the Third Annual Joint Conference of the International Network on Personal Relationships, Prescott, AZ.
- Bartel, C. A. 2001. Social comparisons in boundary-spanning work: Effects of community outreach on members' organizational identity and identification. **Administrative Science Quarterly**, 46: 379–413.
- Becker, T. E. 2005. Potential problems in the statistical control of variables in organizational research: A qualitative analysis with recommendations. **Organizational Research Methods**, 8: 274–289.
- Beechler, S., S ndergaard, M., Miller, E. L., & Bird, A. 2004. Boundary spanning. In H. W. Lane, M. L. Maznevski, M. E. Mendenhall & J. McNett (Eds.), **The Blackwell handbook of global management: A guide to managing complexity:** 121–133. Oxford, UK: Blackwell Publishing Ltd.
- Beyer, J. M., Chattopadhyay, P., George, E., Glick, W. H., and Ogilvie, D. T., & Pugliese, D. 1997. The selective perception of managers revisited. **Academy of Management Journal**, 40: 716–737.
- Bieri, J. 1955. Cognitive complexity-simplicity and predictive behavior. **Journal of Abnormal Psychology**, 51: 263–268.
- Blau, P. M. 1977. **Inequality and heterogeneity: A primitive theory of social structure.** New York, NY: Free Press.
- Bouty, I. 2000. Interpersonal and interaction influences on informal resource exchanges between R&D researchers across organizational boundaries. **Academy of Management Journal**, 43: 50–65.
- Buckenham, M. A. 1998. Socialization and personal change: A personal construct psychology approach. **Journal of Advanced Nursing**, 28: 874–881.
- Bunderson, J. S. 2003. Team member functional background and involvement in management teams: Direct effects and the moderating role of power centralization. **Academy of Management Journal**, 46: 458–474.
- Bunderson, J. S., & Sutcliffe, K. M. 2002. Comparing alternative conceptualizations of functional diversity in management teams: Process and performance effects. **Academy of Management Journal**, 45: 875–893.
- Burke, L. A., & Steensma, H. K. 1998. Towards a model for relating executive career experiences and firm performance. **Journal of Managerial Issues**, 10: 86–102.
- Burleson, B. R. 2007. Constructivism: A general theory of communication skill. In B. B. Whaley & W. Samter (Eds.), **Explaining communication: Contemporary theories and exemplars:** 105–128. Mahwah, NJ: Lawrence Erlbaum Associates.
- Burleson, B. R., & Caplan, S. E. 1998. Cognitive complexity. In J. C. McCroskey, J. A. Daly, M. M. Martin & M. J. Beatty (Eds.), **Communication and personality: Trait perspectives:** 233–286. Cresskill, NJ: Hampton Press.
- Burleson, B. R., & Denton, W. H. 1997. The relationship between communication skill and marital satisfaction: Some moderating effects. **Journal of Marriage and the Family**, 59: 884–902.
- Burleson, B. R., & Samter, W. 1990. Effects of cognitive complexity on the perceived importance of communication skills in friends. **Communication Research**, 17: 165–182.
- Burleson, B. R., & Waltman, M. S. 1988. Cognitive complexity: Using the Role Category Questionnaire mea-

- sure. In C. H. Tardy (Ed.), *A handbook for the study of human communication: Methods and instruments for observing, measuring, and assessing communication processes*: 1–35. Norwood, NJ: Ablex.
- Buyl, T., Boone, C., Hendriks, W., & Matthyssens, P. 2011. Top management team functional diversity and firm performance: The moderating role of CEO characteristics. *Journal of Management Studies*, 48: 151–177.
- Chimhanzi, J. 2004. The impact of integration mechanisms on marketing/HR dynamics. *Journal of Marketing Management*, 20: 713–740.
- Choi, J. N. 2002. External activities and team effectiveness: Review and theoretical development. *Small Group Research*, 33: 181–208.
- Christ, O., van Dick, R., Wagner, U., & Stellmacher, J. 2003. When teachers go the extra mile: Foci of organisational identification as determinants of different forms of organisational citizenship behaviour among schoolteachers. *British Journal of Educational Psychology*, 73: 329–341.
- Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. 2003. *Applied multiple regression/correlation analysis for the behavioral sciences*. New York, NY: Lawrence Erlbaum Associates.
- Cortina, J. M. 1993. What is coefficient alpha? An examination of theory and applications. *Journal of Applied Psychology*, 78: 98–104.
- Crockett, W. H. 1965. Cognitive complexity and impression formation. In B. A. Maher (Ed.), *Progress in experimental personality research* (2nd ed.): 47–90. New York, NY: Academic Press.
- Crockett, W. H., Press, A. N., Delia, J. G., & Kenny, C. T. 1974. The structural analysis of the organization of written impressions. Unpublished manuscript, Department of Psychology, University of Kansas, Lawrence, KS.
- Cycyota, C. S., & Harrison, D. A. 2006. What (not) to expect when surveying executives: A meta-analysis of top manager response rates and techniques over time. *Organizational Research Methods*, 9: 133–160.
- Davison, R. B., Hollenbeck, J. R., Barnes, C. M., Slesman, D. J., & Ilgen, D. R. 2012. Coordinated action in multiteam systems. *Journal of Applied Psychology*, 97: 808–824.
- Dawes, P. L., & Massey, G. R. 2005. Antecedents of conflict in marketing's cross-functional relationship with sales. *European Journal of Marketing*, 39: 1327–1344.
- DeChurch, L. A., & Marks, M. A. 2006. Leadership in multiteam systems. *Journal of Applied Psychology*, 91: 311–329.
- Delia, J. G. 1977. Constructivism and the study of human communication. *Quarterly Journal of Speech*, 63: 66–85.
- Delia, J. G., Clark, R. A., & Switzer, D. E. 1979. The content of informal conversations as a function of interactants' interpersonal cognitive complexity. *Communication Monographs*, 46: 274–281.
- Delia, J. G., & Crockett, W. H. 1973. Social schemas, cognitive complexity, and the learning of social structures. *Journal of Personality*, 41: 413–429.
- Denton, W. H., Burleson, B. R., & Sprenkle, D. H. 1995. Association of interpersonal cognitive complexity with communication skill in marriage: Moderating effects of marital distress. *Family Process*, 34: 101–111.
- DeShon, R. P., Kozlowski, S. W. J., Schmidt, A. M., Milner, K. R., & Wiechmann, D. 2004. A multiple-goal, multilevel model of feedback effects on the regulation of individual and team performance. *Journal of Applied Psychology*, 89: 1035–1056.
- Dougherty, D. 1992. Interpretive barriers to successful product innovation in large firms. *Organization Science*, 3: 179–202.
- Drach-Zahavy, A. 2011. Interorganizational teams as boundary spanners: The role of team diversity, boundedness, and extrateam links. *European Journal of Work and Organizational Psychology*, 20: 89–118.
- Drach-Zahavy, A., & Somech, A. 2010. From an intrateam to an interteam perspective of effectiveness: The role of interdependence and boundary activities. *Small Group Research*, 41: 143–174.
- Dunn, W. N., & Ginsberg, A. 1986. A sociocognitive network approach to organizational analysis. *Human Relations*, 39: 955–975.
- Dutton, J. E., Dukerich, J. M., & Harquail, C. V. 1994. Organizational images and member identification. *Administrative Science Quarterly*, 39: 239–263.
- Duys, D. K., & Hedstrom, S. M. 2000. Basic counselor skills training and counselor cognitive complexity. *Counselor Education and Supervision*, 40: 8–18.
- Edwards, J. R., & Lambert, L. S. 2007. Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychological Methods*, 12: 1–22.
- Ellemers, N., de Gilder, D., & Haslam, S. A. 2004. Motivating individuals and groups at work: A social identity perspective on leader and group performance. *Academy of Management Review*, 29: 459–478.
- Ellemers, N., & Rink, F. 2005. Identity in work groups:

- The beneficial and detrimental consequences of multiple identities and group norms for collaboration and group performance. *Advances in Group Processes*, 22: 1–41.
- Evans, M. G. 1985. A Monte Carlo study of the effects of correlated method variance in moderated multiple regression analysis. *Organizational Behavior and Human Decision Processes*, 36: 305–323.
- Faraj, S., & Yan, A. 2009. Boundary work in knowledge teams. *Journal of Applied Psychology*, 94: 604–617.
- Fisher, R. J., Maltz, E., & Jaworski, B. J. 1997. Enhancing communication between marketing and engineering: The moderating role of relative functional identification. *Journal of Marketing*, 61: 54–70.
- Fournier, V. 1996. Cognitive maps in the analysis of personal change during work role transitions. *British Journal of Management*, 7: 87–105.
- Fournier, V., & Payne, R. 1994. Change in self construction during the transition from university to employment: A personal construct psychology approach. *Journal of Occupational and Organizational Psychology*, 67: 297–314.
- Fransella, F. (Ed.). 2003. *International handbook of personal construct psychology*. Sussex, UK: John Wiley & Sons, Ltd.
- Friedman, R. A., & Podolny, J. 1992. Differentiation of boundary spanning roles: Labor negotiations and implications for role conflict. *Administrative Science Quarterly*, 37: 28–47.
- Gaertner, S. L., Dovidio, J. F., Banker, B. S., Houlette, M., Johnson, K. M., & McGlynn, E. A. 2000. Reducing intergroup conflict: From superordinate goals to decategorization, recategorization, and mutual differentiation. *Group Dynamics: Theory, Research, and Practice*, 4: 98–114.
- Ginsberg, A. 1989. Construing the business portfolio: A cognitive model of diversification. *Journal of Management Studies*, 24: 417–438.
- Ginsberg, A. 1990. Connecting diversification to performance: A sociocognitive approach. *Academy of Management Journal*, 15: 514–535.
- Goodwin, G. F., Essens, P. J. M. D., & Smith, D. 2012. Multi-team systems in the public sector. In S. J. Zaccaro, M. A. Marks & L. A. DeChurch (Eds.), *Multi-team systems: An organization form for dynamic and complex environments*. Sussex, UK: Psychology Press Ltd.
- Goodwin, V. L., & Ziegler, L. 1998. A test of relationships in a model of organizational cognitive complexity. *Journal of Organizational Behavior*, 19: 371–386.
- Griffin, A., & Hauser, J. R. 1996. Integrating R&D and marketing: A review and analysis of the literature. *Journal of Product Innovation Management*, 13: 191–215.
- Hambrick, D. C., and Mason, P. A. 1984. Upper echelons: The organization as a reflection of its top managers. *Academy of Management Review*, 9: 193–206.
- Hale, C. L. 1980. Cognitive complexity-simplicity as a determinant of communication effectiveness. *Communication Monographs*, 47: 304–311.
- Hale, C. L., & Delia, J. G. 1976. Cognitive complexity and social perspective-taking. *Communication Monographs*, 43: 195–203.
- Hansen, M. T. 2002. Knowledge networks: Explaining effective knowledge sharing in multiunit companies. *Organization Science*, 13: 232–248.
- Hansen, M. T., Mors, M. L., & Løvås, B. 2005. Knowledge sharing in organizations: Multiple networks, multiple phases. *Academy of Management Journal*, 48: 776–793.
- Harrison, D. A., & Klein, K. J. 2007. What's the difference? Diversity constructs as separation, variety, or disparity in organizations. *Academy of Management Review*, 32: 1199–1228.
- Hoegl, M., Weinkauff, K., & Gemuenden, H. G. 2004. Interteam coordination, project commitment, and teamwork in multiteam R&D projects: A longitudinal study. *Organization Science*, 15: 38–55.
- Homburg, C., & Jensen, O. 2007. The thought worlds of marketing and sales: Which differences make a difference? *Journal of Marketing*, 71: 124–142.
- Ibarra, H. 1993. Network centrality, power, and innovation involvement: Determinants of technical and administrative roles. *Academy of Management Journal*, 36: 471–501.
- Joshi, A. 2006. The influence of organizational demography on the external networking behavior of teams. *Academy of Management Review*, 31: 583–595.
- Joshi, A., Pandey, N., & Han, G. H. 2009. Bracketing team boundary spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior*, 30: 731–759.
- Kelly, G. A. 1955. *The psychology of personal constructs*. New York, NY: W. W. Norton.
- Klein, K. J., & Kozlowski, S. W. J. 2000. From micro to meso: Critical steps in conceptualizing and conducting multilevel research. *Organizational Research Methods*, 3: 211–236.
- Kline, S. L. 1991. Construct differentiation and person-centred regulative messages. *Journal of Language and Social Psychology*, 10: 1–27.
- Kline, S. L., & Chatani, K. 2001. Social perception and message awareness as correlates of person-centered

- regulative messages. *Communication Research Reports*, 18: 274–284.
- Kline, S. L., & Floyd, C. H. 1990. On the art of saying no: The influence of social cognitive development on messages of refusal. *Western Journal of Speech Communication*, 54: 454–472.
- Kline, S. L., Pelias, R. J., & Delia, J. G. 1991. The predictive validity of cognitive complexity measures on social perspective-taking and counseling communication. *International Journal of Personal Construct Psychology*, 4: 347–357.
- Lanaj, K., Hollenbeck, J. R., Ilgen, D. R., Barnes, C. M., & Harmon, S. J. 2013. The double-edged sword of decentralized planning in multiteam systems. *Academy of Management Journal*, 56: 735–757.
- Lawrence, P. R., & Lorsch, J. W. 1967. Differentiation and integration in complex organizations. *Administrative Science Quarterly*, 12: 1–47.
- Leenders, M. A. A. M., & Wierenga, B. 2002. The effectiveness of different mechanisms for integrating marketing and R&D. *Journal of Product Innovation Management*, 19: 305–317.
- Leitner, L. M. 1988. Contextual shifts in interpersonal constructions. In F. Fransella & L. Thomas (Eds.), *Experimenting with personal construct psychology*: 308–318. London, UK: Routledge.
- Levina, N., & Vaast, E. 2005. The emergence of boundary spanning competence in practice: Implications for implementation and use of information systems. *Management Systems Quarterly*, 29: 335–363.
- Louis, M. R. 1980. Surprise and sense making: What newcomers experience in entering unfamiliar organizational settings. *Administrative Science Quarterly*, 25: 226–251.
- Mabe, P. A., & West, S. G. 1982. Validity of self-evaluation of ability: A review and meta-analysis. *Journal of Applied Psychology*, 67: 280–296.
- Mael, F. A., & Ashforth, B. E. 1992. Alumni and their alma mater: A partial test of the reformulated model of organizational identification. *Journal of Organizational Behavior*, 13: 103–123.
- Maltz, E. 1997. An enhanced framework for improving cooperation between marketing and other functions: The differential role of integrating mechanisms. *Journal of Market Focused Management*, 2: 83–98.
- Maltz, E., & Kohli, A. K. 2000. Reducing marketing's conflict with other functions: The differential effects of integrating mechanisms. *Journal of the Academy of Marketing Science*, 28: 479–492.
- Markóczy, L. 1997. Measuring beliefs: Accept no substitutes. *Academy of Management Journal*, 40: 1228–1242.
- Marks, M. A., DeChurch, L. A., Mathieu, J. E., Panzer, F. J., & Alonso, A. 2005. Teamwork in multiteam systems. *Journal of Applied Psychology*, 90: 964–971.
- Marrone, J. A. 2004. Cutting across team boundaries: Antecedents and implications of individual boundary spanning behavior within consulting teams. Doctoral dissertation, University of Maryland, College Park, Maryland.
- Marrone, J. A. 2010. Team boundary spanning: A multi-level review of past research and proposals for the future. *Journal of Management*, 36: 911–940.
- Marrone, J. A., Tesluk, P. E., & Carson, J. B. 2007. A multilevel investigation of antecedents and consequences of team member boundary-spanning behavior. *Academy of Management Journal*, 50: 1423–1440.
- Mathieu, J. E., Marks, M. A., & Zaccaro, S. J. 2001. Multi-team systems. In N. Anderson, D. S. Ones, H. K. Sinangil & C. Viswesvaran (Eds.), *Handbook of industrial, work and organizational psychology*: 289–313. London, UK: Sage.
- Matusik, S. F., & Hill, C. W. L. 1998. The utilization of contingent work, knowledge creation, and competitive advantage. *Academy of Management Review*, 23: 680–697.
- McCann, J. E., & Ferry, D. L. 1979. An approach for assessing and managing inter-unit interdependence. *Academy of Management Review*, 4: 113–119.
- McGuire, M. 2006. Collaborative public management: Assessing what we know and how we know it. *Public Administration Review*, 66: 33–43.
- Miles, R. H., & Perreault, W. D. 1976. Organizational role conflict: Its antecedents and consequences. *Organizational Behavior and Human Performance*, 17: 19–44.
- Nunnally, J. C. 1978. *Psychometric theory* (2nd ed.). New York, NY: McGraw-Hill.
- O'Keefe, B. J., & Delia, J. G. 1979. Construct comprehensiveness and cognitive complexity as predictors of the number and strategic adaptation of arguments and appeals in a persuasive message. *Communication Monographs*, 46: 231–240.
- O'Keefe, B. J., & Shepherd, G. J. 1987. The pursuit of multiple objectives in face-to-face persuasive interactions: Effects of construct differentiation on message organization. *Communication Monographs*, 54: 396–419.
- O'Keefe, D. J., Shepherd, G. J., & Streeter, T. 1982. Role category questionnaire measures of cognitive complexity: Reliability and comparability of alternative forms. *Central States Speech Journal*, 33: 333–338.
- O'Keefe, D. J., & Sypher, H. E. 1981. Cognitive complexity

- measures and the relationship of cognitive complexity to communication. *Human Communication Research*, 8: 72–92.
- O'Leary, M. B., Mortensen, M., & Woolley, A. W. 2011. Multiple team membership: A theoretical model of its effects on productivity and learning for individuals and teams. *Academy of Management Review*, 36: 461–478.
- Parker, C. P., Dipboye, R. L., & Jackson, S. L. 1995. Perceptions of organizational politics: An investigation of antecedents and consequences. *Journal of Management*, 21: 891–912.
- Parker, S. K., & Axtell, C. M. 2001. Seeing another viewpoint: Antecedents and outcomes of employee perspective taking. *Academy of Management Journal*, 44: 1085–1100.
- Parry, M. E., & Song, X. M. 1993. Determinants of R&D–marketing integration in high-tech Japanese firms. *Journal of Product Innovation Management*, 10: 4–22.
- Penley, L. E., Alexander, E. R., Jernigan, I. E., & Henwood, C. I. 1991. Communication abilities of managers: The relationship to performance. *Journal of Management*, 17: 57–76.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. 2003. Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88: 879–903.
- Prahalad, C. K., & Bettis, R. A. 1986. The dominant logic: A new linkage between diversity and performance. *Strategic Management Journal*, 7: 485–501.
- Ramarajan, L., Bezrukova, K., Jehn, K. A., & Euwema, M. 2011. From the outside in: The negative spillover effects of boundary spanners' relations with members of other organizations. *Journal of Organizational Behavior*, 32: 886–905.
- Raskas, D. F., & Brick, D. C. 1992. Multifunctional managerial development: A framework for evaluating the options. *Organizational Dynamics*, 21: 5–17.
- Richter, A. W., West, M. A., Van Dick, R., & Dawson, J. F. 2006. Boundary spanners' identification, intergroup contact, and effective intergroup relations. *Academy of Management Journal*, 49: 1252–1269.
- Riketta, M., & Van Dick, R. 2005. Foci of attachment in organizations: A meta-analytic comparison of the strength and correlates of workgroup versus organizational identification and commitment. *Journal of Vocational Behavior*, 67: 490–510.
- Scarlett, H. H., Press, A. N., & Crockett, W. H. 1971. Children's descriptions of peers: A Wernerian developmental analysis. *Child Development*, 42: 439–453.
- Shrout, P. E., & Bolger, N. 2002. Mediation in experimental and nonexperimental studies: New procedures and recommendations. *Psychological Methods*, 7: 422–445.
- Singh, J., Goolsby, J. R., & Rhoads, G. K. 1994. Behavioral and psychological consequences of boundary spanning burnout for customer service representatives. *Journal of Marketing Research*, 31: 558–569.
- Snijders, T. A. B., & Bosker, R. J. 1999. *Multilevel analysis: An introduction to basic and advanced multilevel modeling*. London, UK: Sage.
- Smith, K. G., Carroll, S. J., & Ashford, S. J. 1995. Intra- and interorganizational cooperation: Toward a research agenda. *Academy of Management Journal*, 38: 7–23.
- Spector, P. E. 2006. Method variance in organizational research: Truth or urban legend? *Organizational Research Methods*, 9: 221–232.
- Spekman, R. E. 1979. Influence and information: An exploratory investigation of the boundary role person's basis of power. *Academy of Management Journal*, 22: 104–117.
- Suedfeld, P., & Bluck, S. 1993. Changes in integrative complexity accompanying significant life events: Historical evidence. *Journal of Personality and Social Psychology*, 64: 124–130.
- Sypher, B. D., & Zorn, T. E. 1986. Communication-related abilities and upward mobility: A longitudinal investigation. *Human Communication Research*, 12: 420–431.
- Sypher, H. E., & Sypher, B. D. 1988. Cognitive differentiation and communication behavior: The Role Category Questionnaire. *Management Communication Quarterly*, 2: 283–294.
- Tajfel, H. 1982. Social psychology of intergroup relations. *Annual Review of Psychology*, 33: 1–39.
- Tsai, W. 2002. Social structure of “coopetition” within a multiunit organization: Coordination, competition, and intraorganizational knowledge sharing. *Organization Science*, 13: 179–190.
- Turner, J. C., Hogg, M. A., Oakes, P. J., Reicher, S. D., & Wetherell, M. S. 1987. *Rediscovering the social group: A self-categorization theory*. Oxford, UK: Basil Blackwell.
- Turner, L. 2008. Attachment style of children in foster care and its relationship to interpersonal cognitive complexity. *Public Administration*, Doctoral dissertation, Wichita State University, Wichita, Kansas.
- Ullrich, J., Wieseke, J., Christ, O., Schulze, M., & Van Dick, R. 2007. The identity-matching principle: Corporate and organizational identification in a franchising system. *British Journal of Management*, 18: S29–S44.

- van der Vegt, G. S., van de Vliert, E., & Oosterhof, A. 2003. Informational dissimilarity and organizational citizenship behavior: The role of intrateam interdependence and team identification. *Academy of Management Journal*, 46: 715–727.
- Walker, B. M., & Winter, D. A. 2007. The elaboration of personal construct psychology. *Annual Review of Psychology*, 58: 453–477.
- Waller, M. J., Huber, G. P., & Glick, W. H. 1995. Functional background as a determinant of executives' selective perception. *Academy of Management Journal*, 38: 943–974.
- Wasserman, S., & Faust, K. 1994. *Social network analysis* (18th ed.). Cambridge, UK: Cambridge University Press.
- Wicker, A. 1969. Cognitive complexity, school size, and participation in school behavior settings: A test of the frequency of interaction hypothesis. *Journal of Educational Psychology*, 60: 200–203.
- Williams, P. 2002. The competent boundary spanner. *Public Administration*, 80: 103–124.
- Wilson, S. R. 1990. Development and test of a cognitive rules model of interaction goals. *Communication Monographs*, 57: 81–103.
- Woods, E. 1996. Associations of nonverbal decoding ability with indices of person-centered communicative ability. *Communication Reports*, 9: 13–22.
- Zhang, M., Xin, Z., & Lin, C. 2012. Measures of cognitive complexity and its development in Chinese adolescents. *Journal of Constructivist Psychology*, 25: 91–111.
- Zinkhan, G. M., & Braunsberger, K. 2004. The complexity of consumers' cognitive structures and its relevance to consumer behavior. *Journal of Business Research*, 57: 575–582.
- Zorn, T. E., & Violanti, M. T. 1996. Communication abilities and individual achievement in organizations. *Management Communications Quarterly*, 10: 139–167.



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